

Calculul matricii de transfer a celulei elementare

$$M_{cE} = \frac{1}{4ik\alpha} \begin{pmatrix} (\alpha+ik)e^{-\alpha b} & -(\alpha+ik)e^{\alpha b} \\ -(\alpha+ik)e^{-\alpha b} & (\alpha+ik)e^{\alpha b} \end{pmatrix} \begin{pmatrix} \alpha+ik & \alpha-ik \\ \alpha-ik & \alpha+ik \end{pmatrix} \begin{pmatrix} e^{ika} & 0 \\ 0 & e^{-ika} \end{pmatrix}$$

$$\begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix}$$

$$c_{11} = (\alpha+ik)^2 e^{-\alpha b} - (\alpha-ik)^2 e^{\alpha b} = (\alpha^2 + 2ik\alpha - k^2) e^{-\alpha b} - (\alpha^2 - 2ik\alpha - k^2) e^{\alpha b} = (\alpha^2 - k^2) (e^{-\alpha b} - e^{\alpha b}) + 2i\alpha k (e^{-\alpha b} + e^{\alpha b}) = -(\alpha^2 - k^2) 2 \sinh(\alpha b) + 2i\alpha k 2 \cosh(\alpha b)$$

$$c_{12} = (\alpha+ik)(\alpha-ik) e^{-\alpha b} - (\alpha-ik)(\alpha+ik) e^{\alpha b} = (\alpha^2 + k^2) e^{-\alpha b} - (\alpha^2 + k^2) e^{\alpha b} = -(\alpha^2 + k^2) (e^{-\alpha b} - e^{\alpha b}) = -(\alpha^2 + k^2) 2 \sinh(\alpha b)$$

$$c_{21} = -(\alpha-ik)(\alpha+ik) e^{-\alpha b} + (\alpha+ik)(\alpha-ik) e^{\alpha b} = -(\alpha^2 + k^2) e^{-\alpha b} + (\alpha^2 + k^2) e^{\alpha b} = (\alpha^2 + k^2) (e^{\alpha b} - e^{-\alpha b}) = (\alpha^2 + k^2) 2 \sinh(\alpha b)$$

$$c_{22} = -(\alpha-ik)^2 e^{-\alpha b} + (\alpha+ik)^2 e^{\alpha b} = -(\alpha^2 - 2ik\alpha - k^2) e^{-\alpha b} + (\alpha^2 + 2ik\alpha - k^2) e^{\alpha b} = (\alpha^2 - k^2) 2 \sinh(\alpha b) + 2i\alpha k 2 \cosh(\alpha b)$$

$$M_{cE} = \frac{1}{4ik\alpha} \begin{pmatrix} -(\alpha^2 - k^2) 2 \sinh(\alpha b) + 4ik\alpha \cosh(\alpha b) & -2(\alpha^2 + k^2) \sinh(\alpha b) \\ 2(\alpha^2 + k^2) \sinh(\alpha b) & 2(\alpha^2 - k^2) \sinh(\alpha b) + 4ik\alpha \cosh(\alpha b) \end{pmatrix} \begin{pmatrix} e^{ika} & 0 \\ 0 & e^{-ika} \end{pmatrix}$$

$$M_{CE} = \begin{pmatrix} -\frac{\alpha^2 - k^2}{2ik\alpha} \sinh(\alpha b) + \cosh(\alpha b) & \frac{\alpha^2 + k^2}{2ik\alpha} \sinh(\alpha b) \\ \frac{\alpha^2 + k^2}{2ik\alpha} \sinh(\alpha b) & \frac{\alpha^2 - k^2}{2ik\alpha} \sinh(\alpha b) + \cosh(\alpha b) \end{pmatrix} \begin{pmatrix} e^{-ika} & 0 \\ 0 & e^{ika} \end{pmatrix}$$

$$M_{CE} = \begin{pmatrix} \frac{\alpha^2 - k^2}{2k\alpha} i \sinh(\alpha b) + \cosh(\alpha b) & \frac{\alpha^2 + k^2}{2ik\alpha} \sinh(\alpha b) \\ \frac{\alpha^2 + k^2}{2ik\alpha} \sinh(\alpha b) & -\frac{\alpha^2 - k^2}{2k\alpha} i \sinh(\alpha b) + \cosh(\alpha b) \end{pmatrix} \begin{pmatrix} e^{-ika} & 0 \\ 0 & e^{ika} \end{pmatrix}$$

$$\cosh(x) = \frac{e^x + e^{-x}}{2} \quad \sinh(x) = \frac{e^x - e^{-x}}{2}$$

$$\cosh(x) = \cos(ix) \quad \sinh(x) = -i \sin(ix)$$

$$M_{CE} = \begin{pmatrix} \left[\frac{\alpha^2 - k^2}{2k\alpha} i \sinh(\alpha b) + \cosh(\alpha b) \right] e^{-ika} & \frac{\alpha^2 + k^2}{2ik\alpha} \sinh(\alpha b) e^{ika} \\ \frac{\alpha^2 + k^2}{2ik\alpha} \sinh(\alpha b) e^{-ika} & \left[-\frac{\alpha^2 - k^2}{2k\alpha} i \sinh(\alpha b) + \cosh(\alpha b) \right] e^{ika} \end{pmatrix}$$